

Cambridge (CIE) IGCSE Chemistry
Paper 4: Theory (Set B)
Extended

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

/ 80

Instructions

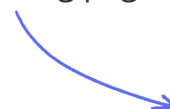
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- You should show all your working and use appropriate units.

Materials

- A copy of [the Periodic Table of elements](#) is required for this paper.

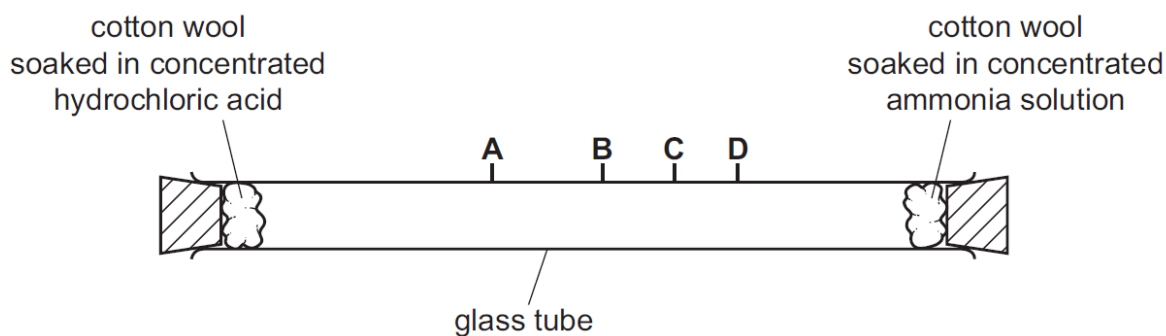
Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 (a)** Concentrated ammonia solution gives off ammonia gas. Concentrated hydrochloric acid gives off hydrogen chloride gas. Ammonia, NH_3 , and hydrogen chloride, HCl , are both colourless gases. Ammonia reacts with hydrogen chloride to make the white solid ammonium chloride.

Apparatus is set up as shown.



After ten minutes a white solid forms in the tube where the gases meet.

- i) Write the chemical equation for the reaction of ammonia with hydrogen chloride.

[1]

- ii) Name the process by which the ammonia and hydrogen chloride gases move in the tube.

[1]

- iii) At which point, A, B, C or D, does the white solid form? Explain why the white solid forms at that point.

[3]

- iv) The experiment was repeated at a higher temperature. Predict how the results of the experiment would be different. Explain your answer.

[3]

.....

.....

.....

.....

.....

(8 marks)

(b) Some of the white solid is removed from the tube and dissolved in water.

Describe how the white solid could be tested to show it contains:

i) Ammonium ions.

Test

Result

[3]

ii) Chloride ions.

Test

Result

[3]

.....

.....

.....

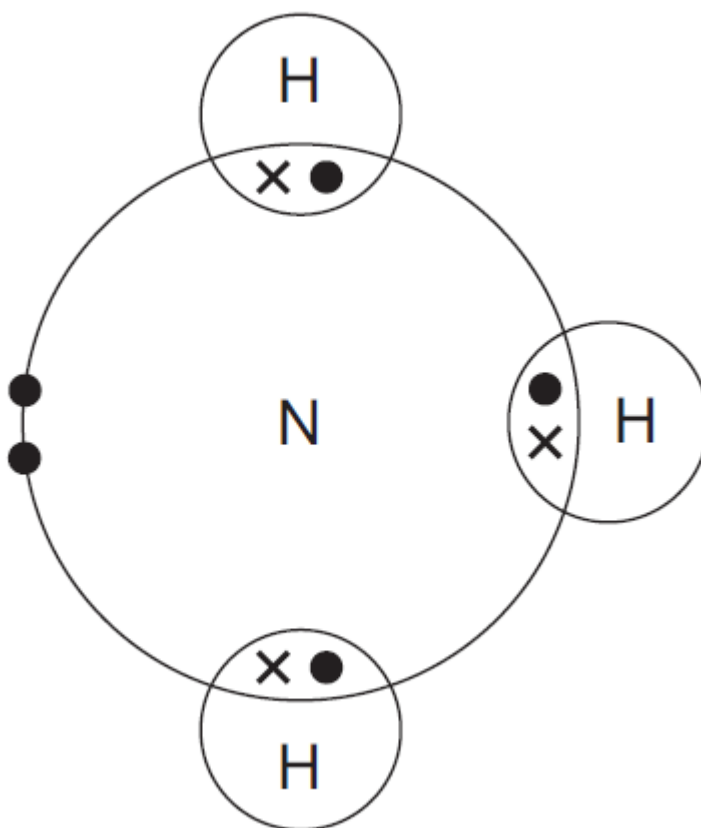
.....

.....

.....

(6 marks)

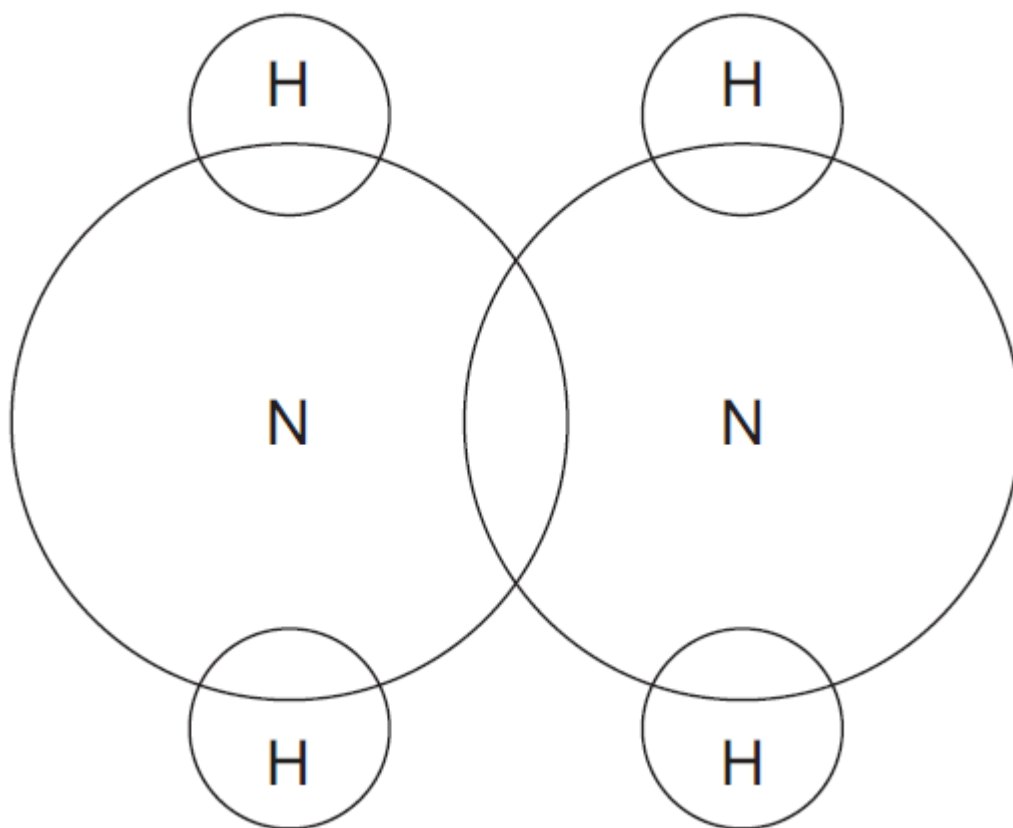
- (c) The diagram shows the electron arrangement in a molecule of ammonia, showing only outer shell electrons.



- i) State the type of bonding in ammonia.

[1]

ii) Hydrazine, N_2H_4 , is another compound of nitrogen and hydrogen. Complete the diagram to show the electron arrangement in a molecule of hydrazine, showing only outer shell electrons.



[3]

.....

.....

.....

.....

(4 marks)

(d) Nylon and proteins are both polymers containing nitrogen.

i) Name the linkages found in the polymers of nylon and protein.

[1]

ii) Describe one difference in the structures of nylon and protein.

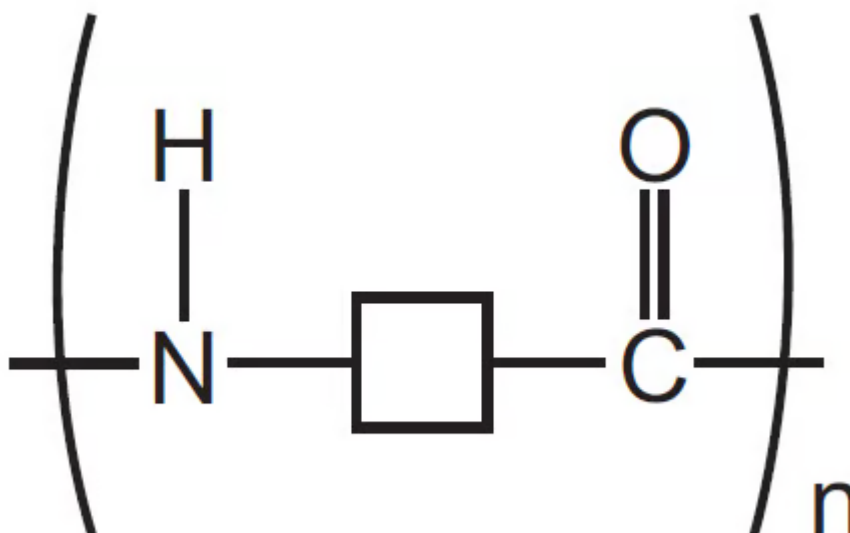
[1]

iii) What is the general name given to the products of hydrolysis of proteins?

[1]

(3 marks)

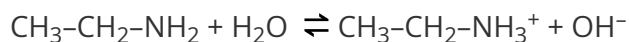
(e) Suggest the structure of the monomer used to make the polymer shown.



(1 mark)

- 2 (a)** Ethylamine, $\text{CH}_3\text{-CH}_2\text{-NH}_2$, is a base which has similar properties to ammonia.

In aqueous ethylamine, there is the following equilibrium.



Explain why water is behaving as an acid in this reaction.

.....
(1 mark)

- (b)** Given aqueous solutions of ethylamine and sodium hydroxide, describe how you could show that ethylamine is a weak base like ammonia and not a strong base like sodium hydroxide.

.....
.....
.....
(3 marks)

- (c)** Ethylamine, like ammonia, reacts with acids to form salts.



Suggest how you could displace ethylamine from the salt, ethylammonium chloride.

.....
.....
(2 marks)

- (d)** Explain the chemistry of the following reaction:

When aqueous ethylamine is added to aqueous iron(III) chloride, a brown precipitate is formed.

(2 marks)

- 3 (a)** The table below shows the elements in the second period of the Periodic Table and some of their oxidation states in their most common compounds.

element	Li	Be	B	C	N	O	F	Ne
number of outer electrons	1	2						
oxidation state	+1	+2	+3	+4	-3	-2	-1	0

i) What does it mean when the only oxidation state of an element is zero?

[1]

ii) Explain why some elements have positive oxidation states but others have negative ones.

[2]

iii) Select two elements in the table which exist as diatomic molecules of the type X_2 .

[2]

.....

.....

.....

.....

.....

(5 marks)

(b) Beryllium hydroxide, a white solid, is an amphoteric hydroxide.

i) Name another metal which has an amphoteric hydroxide.

[1]

ii) Suggest what you would observe when an excess of aqueous sodium hydroxide is added gradually to aqueous beryllium sulfate.

[2]

.....

.....

.....

(3 marks)

4 (a) Aqueous sodium hydroxide is a base.

Complete this sentence about the different types of bases. Bases are metal hydroxides or metal _____

.....
(1 mark)

(b) Describe the reaction of aqueous sodium hydroxide with:

i) A named acid

ii) An ammonium salt

.....
.....
.....
.....
(4 marks)

(c) Ammonia is a soluble base.

Draw a circle around the pH value of aqueous ammonia.

pH 1 pH 5 pH 7 pH 10

.....
(1 mark)

- 5 (a)** Manganese is a transition element. It has more than one valency and the metal and its compounds are catalysts.

Predict three other properties of manganese that are typical of transition elements.

(3 marks)

- (b)** It has several oxides, three of which are shown below. Manganese(II) oxide, which is basic. Manganese(III) oxide, which is amphoteric. Manganese(IV) oxide, which is acidic.

i) Complete the word equation.

manganese(II) oxide + hydrochloric acid → +

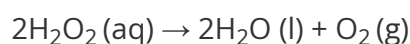
[2]

ii) Which, if any, of these oxides will react with sodium hydroxide?

[1]

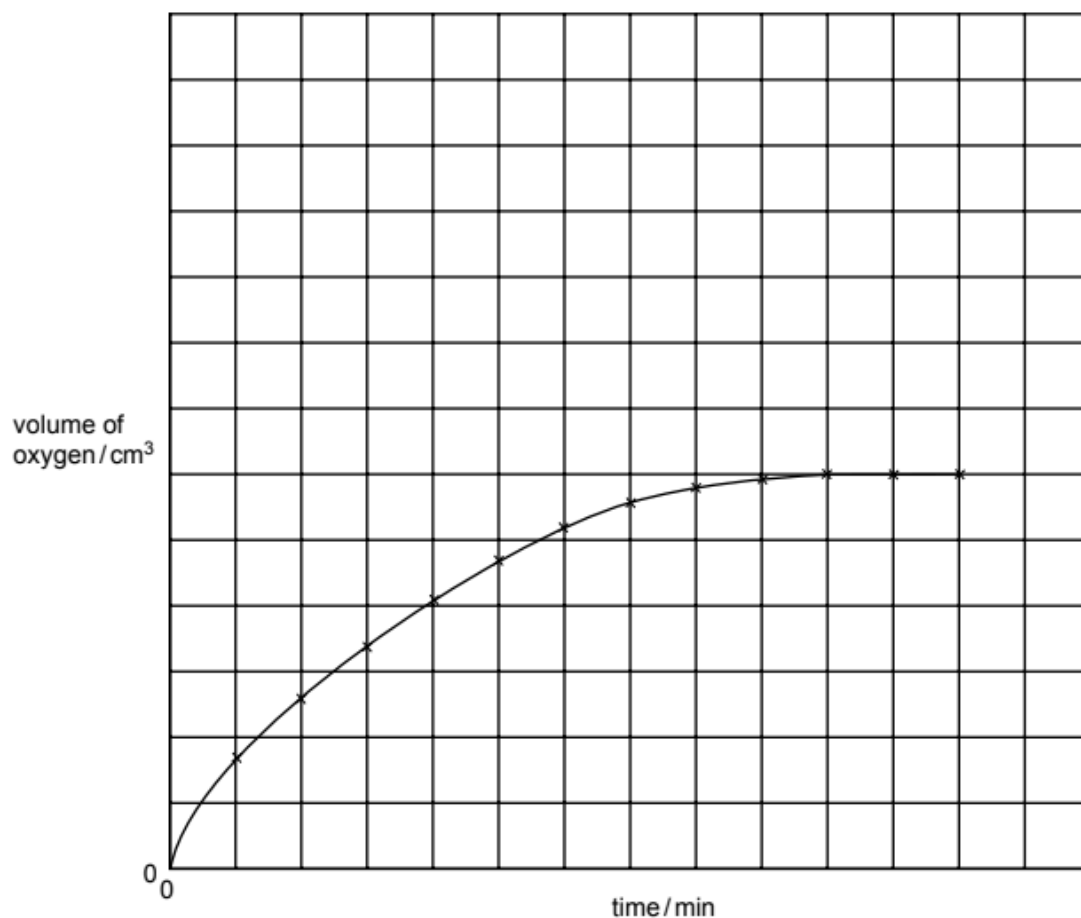
(3 marks)

- (c)** Aqueous hydrogen peroxide decomposes to form water and oxygen.



This reaction is catalysed by manganese(IV) oxide. The following experiments were carried out to investigate the rate of this reaction. A 0.1 g sample of manganese(IV) oxide was added to 20 cm³ of 0.2 mol / dm³ hydrogen peroxide solution. The volume of oxygen

produced was measured every minute. The results of this experiment are shown on the graph.



i) How does the rate of reaction vary with time? Explain why the rate varies.

[3]

ii) The following experiment was carried out at the same temperature. 0.1 g of manganese(IV) oxide and 20 cm³ of 0.4 mol / dm³ hydrogen peroxide. Sketch the curve for this experiment on the same grid.

[2]

iii) How would the shape of the graph differ if only half the mass of catalyst had been used in these experiments?

[2]

.....

.....

.....

.....

.....

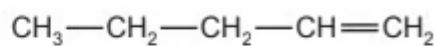
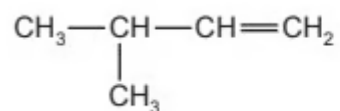
.....

(7 marks)

- 6 (a)** The alkenes are unsaturated hydrocarbons. They form a homologous series, the members of which have the same chemical properties.

They undergo addition reactions and are easily oxidised.

The following hydrocarbons are isomers.



- i) Explain why these two hydrocarbons are isomers.

[2]

- ii) Give the structural formula of another hydrocarbon which is isomeric with the above.

[1]

(3 marks)

(b) Give the structural formula and name of each of the products of the following addition reactions.

i) ethene and bromine structural formula of product

name of product

[2]

ii) propene and hydrogen structural formula of product

name of product

[2]

iii) but-1-ene and water structural formula of product

name of product

[2]

.....

.....

.....

.....

.....

.....

(6 marks)

(c) Alkenes can be oxidised to carboxylic acids.

i) For example, propene, $\text{CH}_3-\text{CH}=\text{CH}_2$, would produce ethanoic acid, CH_3-COOH , and methanoic acid, $\text{H}-\text{COOH}$.

Deduce the formulae of the alkenes which would form the following carboxylic acids when oxidised.

ethanoic acid and propanoic acid

only ethanoic acid

[2]

ii) Describe the colour change you would observe when an alkene is oxidised with acidified potassium manganate(VII).

[2]

.....

.....

.....

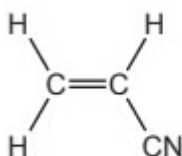
.....

(4 marks)

(d) Alkenes polymerise to form addition polymers.

Draw the structural formula of poly(cyanoethene), include at least two monomer units.

The structural formula of the monomer, cyanoethene, is given below.



(2 marks)

7 (a) There are two types of polymerisation, addition and condensation.

Explain the difference between these two types of polymerisation.

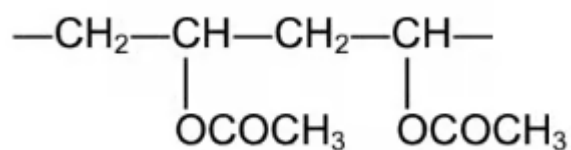
(2 marks)

(b) Some plastics, formed by polymerisation, are non-biodegradable.

Describe **two** pollution problems that are caused by non-biodegradable plastics.

(2 marks)

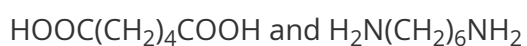
(c) The polymer known as PVA is used in paints and adhesives. Its structural formula is shown below



Deduce the structural formula of its monomer.

(1 mark)

(d) A condensation polymer can be made from the following monomers.



Draw the structural formula of this polymer.

(3 marks)